# The ABCs of Social Network Analysis

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**Abstract**

This document is a very brief note on the social network analysis (SNA). It introduces a few of the most fundamental concepts of SNA, then explains the standard SNA workflow through a quite interesting paper. This is the first document of this series, I plan on updating the note with programming details and advanced techniques. Please note that this document is only suitable for those who know literally nothing about SNA, if you already know what it is, you should save your time and search for more advanced manuals.

## 1 Basic concepts

### 1.1 The data structure of social network

The two most common social network data structures are the **adjacency matrix** and the **connection table**. The adjacency matrix is a square matrix whose size equals the number of nodes; each entry is 0 or 1, indicating whether the row node and the column node are adjacent. Because the rows and columns of a square adjacency matrix index exactly the same set of nodes, the matrix can express directed relationships only through a convention: by convention, in a directed graph the entries are interpreted as relations that run from row to column (see Figure 1(a)).

By contrast, a connection table consists of only two columns: a source node and a target node, with the nodes identified by numeric IDs. Each row of the table therefore records a single directed link. A third column may be added to store the strength of the connection, which is known as the **edge weight** (see Figure 1(b)).

### 1.2 Node importance (centrality)

Social network analysis typically measures the importance of a node within the whole network using **Degree Centrality** and **Closeness Centrality**. In addition, **Betweenness Centrality** captures the extent to which other nodes depend on a given node, and **Eigenvector Centrality** gauges a node's importance through the importance of its neighbors.

#### 1.2.1 Degree centrality

In an undirected graph, the degree centrality of node $i$ is the number of nodes adjacent to it:

$$DC(i) = \sum_j m(i,j)$$

In a directed graph, degree centrality splits further into **in-degree** and **out-degree**.

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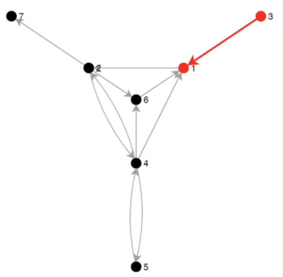
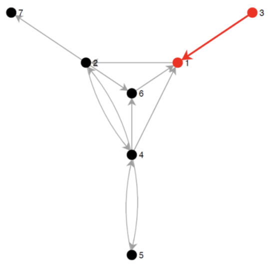



| | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 2 | 0 | 0 | 0 | 1 | 0 | 1 | 1 |
| 3 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 4 | 1 | 1 | 0 | 0 | 1 | 1 | 0 |
| 5 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 6 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 7 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

(a) Adjacency matrix




| | ego | alter |
|---|---|---|
| 1 | 1 | 2 |
| 2 | 2 | 4 |
| 3 | 2 | 6 |
| 4 | 2 | 7 |
| 5 | 3 | 1 |
| 6 | 4 | 1 |
| 7 | 4 | 2 |
| 8 | 4 | 5 |
| 9 | 4 | 6 |
| 10 | 5 | 4 |
| 11 | 6 | 1 |

(b) Connection table

Figure 1: Data structure of social network

#### 1.2.2 Closeness centrality

Closeness centrality measures how central a node is in the whole network, that is, how far it is from every other node. We therefore need to define the distance between two nodes first. Two nodes are usually connected by more than one path; the distance between them is defined as the **length of the shortest path**, and the distance is $\infty$ if no path connects them. Closeness centrality is then the **reciprocal of the average distance** from node $i$ to all other nodes:

$$CC(i) = \frac{n-1}{\sum_{j \neq i} d_{ij}}$$

#### 1.2.3 Betweenness centrality

Betweenness centrality measures the importance of node $i$ by the proportion of shortest paths between other node pairs that pass through node $i$. It is computed as:

$$BC(i) = \sum_{s \neq i \neq t} \frac{\sigma_{st}(i)}{\sigma_{st}}$$

where $\sigma_{st}$ is the number of shortest paths from node $s$ to node $t$, and $\sigma_{st}(i)$ is the number of those paths that pass through node $i$.

#### 1.2.4 Eigenvector centrality

In its standard form, eigenvector centrality is defined for undirected graphs. It measures the importance of node $i$ through the importance of its adjacent nodes. For example, consider a social network $A$ represented by the following adjacency matrix:

$$A = \begin{bmatrix} - & 1 & 1 & 1 & 0 \\ 1 & - & 1 & 1 & 0 \\ 1 & 1 & - & 0 & 0 \\ 1 & 1 & 0 & - & 1 \\ 0 & 0 & 0 & 1 & - \end{bmatrix}$$

Computing the degree centrality of every node yields the vector $V$:

$$V = \begin{bmatrix} 3 \\ 3 \\ 2 \\ 3 \\ 1 \end{bmatrix}$$

Multiplying $\mathbf{A}$ by $\mathbf{V}$ accumulates the degree centrality of node $i$'s neighbors onto node $i$. Repeating the multiplication with the updated vector $\mathbf{V'}$ further propagates the centrality of neighbors' neighbors onto node $i$. Iterating until the vector converges yields the eigenvector equation $\mathbf{Ax} = \lambda \mathbf{x}$, where $\lambda$ is the associated eigenvalue.

This iterative logic can also be written as:

$$x_v = \frac{1}{\lambda} \sum_{u \in N(v)} x_u$$

where $x_v$ is the eigenvector centrality of node $v$ and $N(v)$ is the set of its adjacent nodes. In other words, the eigenvector centrality of node $v$ equals the sum of its neighbors' scores divided by a constant $\lambda$.

## 2 Programming

Stata, R and Python all provide packages for social network analysis, yet the best supported, most widely used and most convenient option remains the dedicated network analysis software UCINET. Note that UCINET was traditionally available for Windows only, though recent releases have provided a Mac version as well. In Stata, the command `nwinstall, all` installs the full set of network analysis commands; Stata requires the data to be imported and the network structure to be declared separately, as documented in the help files. UCINET offers a graphical user interface; see `https://www.lianxh.cn/details/1599.html`.

## 3 An example research

Horváth et al. (2022) contributed a paper to SSRN titled *Social Network Analysis of Collaboration Patterns among Economists in China Based on Chinese- and English-language Publications*[1]. This research is a good example of a standard social network workflow. By building networks at several levels, the paper analyses the following questions, which illustrate the most suitable scenarios for SNA:

1. "What are the patterns of collaboration among Chinese universities..." ⇒ **Identify the collaboration patterns**

[1]Horváth, Gergely, Wanqi Hu, Ruoxian Jiang and Linjia Zhang, 2022. Social Network Analysis of Collaboration Patterns Among Economists in China Based on Chinese- and English-Language Publications. SSRN Working Paper 4200232.

2. "Are the collaboration networks in the English- and Chinese-language publications different from each other?" $\Rightarrow$ **Analyse the collaboration patterns**

3. "Which are the most important universities in the network?" $\Rightarrow$ **Identify the most important nodes**

The SNA scheme provides the following insights:

1. The English-language collaboration network has become increasingly integrated over time.

2. The university-level network can be described as a core–periphery network with the 985 universities in the core.

3. The individual-level collaboration network resembles a small-world network connected by superstar economists.

## 3.1 The SNA design

### 3.1.1 SN construction

This paper considers two networks:

1. the university network:
   - Node: universities
   - Link: publication collaboration between faculty members
   - Weight: the total amount of collaboration

2. the economist network:
   - Node: authors
   - Link: co-authorship relations
   - Weight: the amount of collaboration

Notably, the paper incorporates the time dimension into the network analysis. It constructs a university-level network for each year, and comparing these networks reveals the dynamic changes in collaboration patterns. For the individual-level networks, it adopts a six-year rolling window to avoid excessively scarce links.

### 3.1.2 Macro characteristics

The paper first measures the connectivity of the whole network; this analysis reveals whether all nodes belong to a single integrated network or are split into several separate smaller networks.

Define a component $C$: "A component $C$ is a subset of nodes in the network ($C \subset N$), all nodes in $C$ are connected by a path. Nodes in two different components are not connected..." The authors then count the number of components in the network, as well as the relative number of nodes in the largest component.

The paper also examines the **small-world effect** of academic collaboration. The small-world effect refers to the phenomenon in which every node in a network can reach any other node through a rather short path. A much more popularized version is the concept of six degrees of separation, the idea that everyone in the world can reach anyone else through only a few (around six) intermediaries. The small-world effect therefore basically measures how closely the nodes of a network are connected. A small-world network must at least satisfy the following conditions:

1. Short average path length
2. High clustering coefficient

#### 3.1.3 Micro characteristics

The paper adopts degree centrality and eigenvector centrality to measure the importance of a single node (e.g. a certain university or economist).

The assortativity is also computed to evaluate whether similar nodes tend to be connected to each other. Following Newman (2003), the assortativity coefficient is computed as:

$$r = \frac{\sum_i e_{ii} - \sum_i a_i b_i}{1 - \sum_i a_i b_i}$$

where $e_{ij}$ is the fraction of edges in the network that connect a node of type $i$ to one of type $j$, with $\sum_{ij} e_{ij} = 1$, $a_i = \sum_j e_{ij}$, and $b_j = \sum_i e_{ij}$.

The authors also adopt the **core–periphery model**. This model divides the nodes into two disjoint sets:

- the core nodes, which form a densely connected knit and are connected both to each other and to the periphery nodes;
- the periphery nodes, which are loosely connected to the core nodes but not to each other.

The authors compute all of the above measures using the `NetworkX` Python package.

### 3.2 Results

It is not necessary to list all the results of the original paper; since this document is only a note, I select a few representative results here.

Figure 2 illustrates the university-level networks in different years; the increasing density reflects the growing collaboration among universities. Green nodes denote the 985 universities, red nodes the 211 universities, and black nodes the remaining universities. The figure clearly shows the central position of the 985 universities and the peripheral position of the other universities.

Table 1 reports the macro-level measures of the university-level English-language collaboration network for representative years (the original Table 1 covers every year from 2010 to 2021). The steady increases in the number of nodes, edges, total collaborations and average degree confirm the growing integration of the network over time, while the short average distance in the largest component foreshadows the small-world property discussed above.

Table 1: Summary statistics of the network measures for English-language publications at the university level (representative years)

| Year | Nodes | Edges (unw.) | Collab. | Avg. deg. (unw.) | Avg. deg. (w.) | Clust. coef. | Comp. | Largest comp. | Avg. dist. comp. |
|---|---|---|---|---|---|---|---|---|---|
| 2010 | 34 | 37 | 85 | 2.176 | 5.000 | 0.111 | 4 | 27 | 3.191 |
| 2013 | 64 | 77 | 175 | 2.406 | 5.469 | 0.155 | 5 | 50 | 3.509 |
| 2016 | 79 | 121 | 291 | 3.063 | 7.367 | 0.144 | 2 | 76 | 3.644 |
| 2019 | 100 | 224 | 600 | 4.480 | 12.000 | 0.121 | 3 | 96 | 3.240 |
| 2021 | 130 | 375 | 1140 | 5.769 | 17.538 | 0.175 | 1 | 130 | 3.215 |

Table 2 lists the top five universities by different network measures (the original Table 2 reports the top ten). Universities in Shanghai and Beijing dominate the rankings

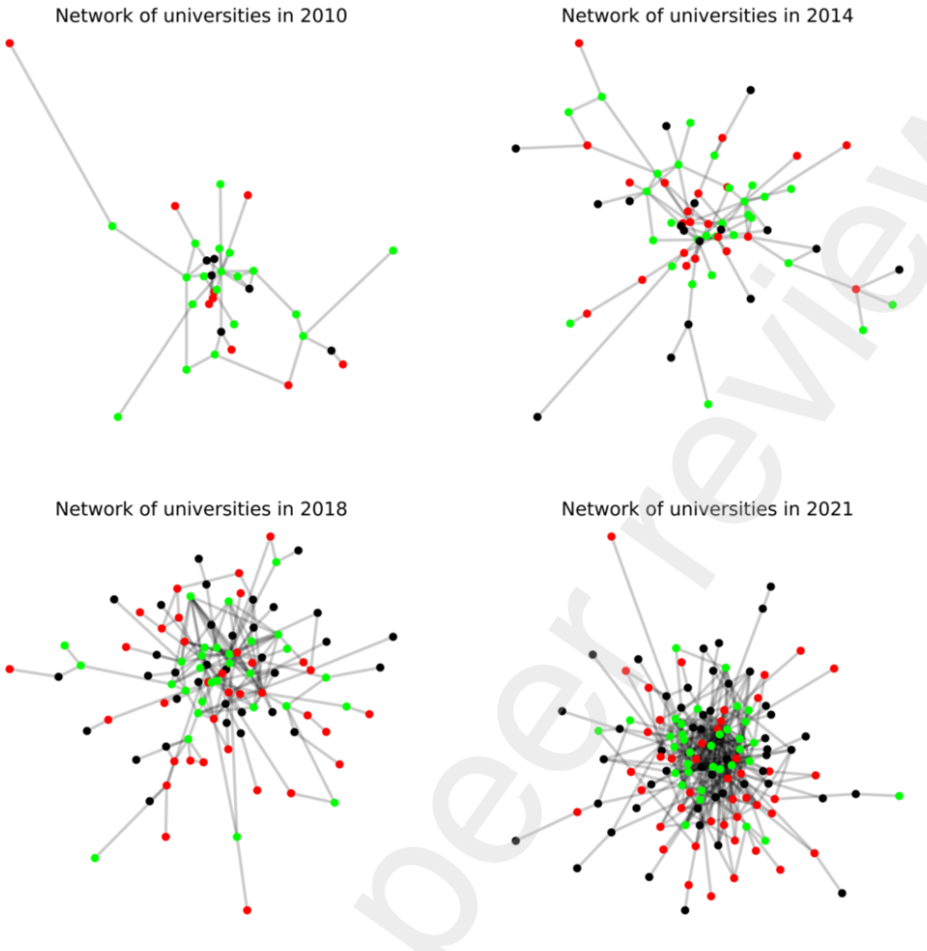



Figure 2: The evolution of the English-language publication network

across all measures: Shanghai University of Finance and Economics leads in degree, Peking University leads in weighted degree and eigenvector centrality, while Huazhong University of Science and Technology ranks first in the coreness index.

Table 2: Top five universities by different network measures

| Rank | Degree | Weighted degree | Eigenvector centrality | Coreness index |
|---|---|---|---|---|
| 1 | Shanghai University of Finance and Economics | Peking University | Peking University | Huazhong University of Science and Technology |
| 2 | Southwestern University of Finance and Economics | Renmin University of China | Shanghai University of Finance and Economics | East China Normal University |
| 3 | Peking University | Shanghai University of Finance and Economics | Renmin University of China | Sun Yat-sen University |
| 4 | Renmin University of China | Central University of Finance and Economics | Southwestern University of Finance and Economics | Southwestern University of Finance and Economics |
| 5 | Zhejiang University | Zhejiang University | Xiamen University | Beijing Institute of Technology |

Table 3 compares the network position of the authors in the top and bottom 20% of the quality distribution. In the English-language publications, the top-quality authors consistently display higher degree and weighted degree than the bottom-quality authors, and this gap widens over time; in the Chinese-language publications, by contrast, the top-quality authors display *lower* degree centrality, reflecting the different nature of the two publication channels.

Table 3: Comparison of network position between top and bottom 20% authors in the quality distribution

**Panel A: English-language publications**

| | 2010–2015 | | 2016–2021 | |
|---|---|---|---|---|
| | Bottom 20% | Top 20% | Bottom 20% | Top 20% |
| Degree | 1.9353 | 2.4797 | 2.2089 | 3.0174 |
| Weighted degree | 2.4641 | 3.2878 | 2.6570 | 4.0116 |
| Clustering | 0.3991 | 0.4956 | 0.4718 | 0.4144 |
| Eigenvector centrality | 0.004 | 0.00001 | 0.0003 | 0.0053 |
| Publication quantity | 2.1533 | 3.7269 | 1.9618 | 3.6991 |
| Publication quality | 0.0000 | 0.9681 | 0.0000 | 0.9204 |

**Panel B: Chinese-language publications**

| | 2010–2015 | | 2016–2021 | |
|---|---|---|---|---|
| | Bottom 20% | Top 20% | Bottom 20% | Top 20% |
| Degree | 2.7216 | 1.7149 | 3.1874 | 1.8019 |
| Weighted degree | 3.9539 | 2.1260 | 4.4435 | 2.2258 |
| Clustering | 0.3352 | 0.2494 | 0.3512 | 0.2842 |
| Eigenvector centrality | 0.0001 | 0.0003 | 0.0003 | 0.0006 |
| Publication quantity | 4.0629 | 2.3136 | 3.9547 | 2.1491 |
| Publication quality | 0.3140 | 1.0000 | 0.3135 | 1.0000 |

Table 4 reports the coefficients from univariate regressions of publication quality on each network measure, with asterisks marking statistical significance. Degree is positively and significantly associated with publication quality in the English-language publications, but the association turns negative in the Chinese-language publications.

Table 4: Regression coefficients from univariate regressions of publication quality on network measures

| | English | | Chinese | |
|---|---|---|---|---|
| Independent variable | 2010–2015 | 2016–2021 | 2010–2015 | 2016–2021 |
| Degree | 0.540*** | 0.795*** | −0.915*** | −1.163*** |
| Eigenvector centrality | −0.004** | 0.005*** | 0.00001 | 0.00001 |
| Clustering | 0.093*** | −0.051*** | −0.121*** | −0.111*** |

Notes: The dependent variable is the author-level publication quality measure. Asterisks denote statistical significance.